

Unit 9, Lesson 2: Density Based on Area



Benchmark

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems that involve density using area.



9.2.1 Warm-Up: Would You Rather?

- Which car would you rather buy?

Car A	Car B
\$19,000	\$25,000
12% depreciation per year when you upgrade the car after 5 years.	16% depreciation per year when you upgrade the car after 5 years.

- Explain your answer.



9.2.2 Collaboration: Population Density

1. The five largest cities in Florida and their populations are shown.

City	Population
Jacksonville	890,467
Miami	454,279
Tampa	387,916
Orlando	280,832
St. Petersburg	261,338

The table shows the area of Florida's five largest cities.

City	Area (square miles)
Jacksonville	747
Miami	35.87
Tampa	113.41
Orlando	102.4
St. Petersburg	61.74

What Do You Notice About the Population and the Area?

2. Discuss with your partner your understanding of population density. Describe the population density for the city or town you live in.

3. Make predictions about the population density for Florida's five largest cities.

Largest Population Density	Smallest Population Density

4. Explain your thinking.

5. Ask a classmate for the predictions they made as well as their reasons. Record their cities and write your summary of their reason. *You are the only person who should write in the middle two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

	City	My Summary of Their Reason	Initials
Largest Population Density			
Smallest Population Density			



To find population density, divide the population by the area.

$$\text{Population Density} = \frac{\text{population}}{\text{area}}$$

6. Determine the population density, in persons per square mile (persons/mi²), of the five largest cities in Florida and record them in the table from largest to smallest. Round answers to the nearest whole number.

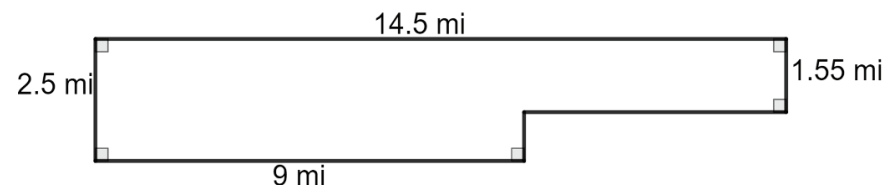
City	Population Density (persons/mi. ²)
1.	
2.	
3.	
4.	
5.	

7. On July 1, 2019, Florida had a population of 21,477,737. The area of Florida is 53,624.76 square miles. Find the population density of Florida.



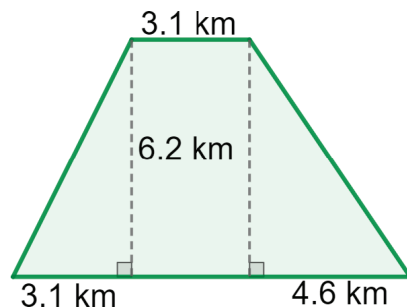
9.2.3 Guided Instruction: Area and Density

1. An outline of the city of Miramar, Florida, with measurements in miles (mi), is shown.



- a. What is the area of the city, in square miles?
- b. Determine the population density of Miramar for 1995 and 2015. Include units.
- | Year | Population | Population Density |
|------|------------|--------------------|
| 1995 | 48,099 | |
| 2015 | 136,996 | |
- c. How did the population density change between 1995 to 2015?

2. The Florida Fish and Wildlife Conservation Commission (FWC) tracks panthers in Florida. By tracking a panther, FWC can map the area in which a panther lives. The habitat of "Florida Panther 251" is shown with dimensions in kilometers, km.



- a. What is the area, in square kilometers (km^2), of Florida Panther 251's habitat?

At one time, the entire southeastern United States was the habitat of the Florida panther. Today, their habitat is restricted to several national parks and wildlife preserves in southwest Florida. Today, the total combined area of the Florida panther habitat is $9,466 \text{ km}^2$.

- b. Based on the area of Florida Panther 251's habitat, how many Florida panthers could be living in southwest Florida?

3. Town A has 20 city blocks, and each city block measures $\frac{1}{20}$ of a mile by $\frac{1}{2}$ mile. There are 6,500 people in this town. Find the population density of the town.



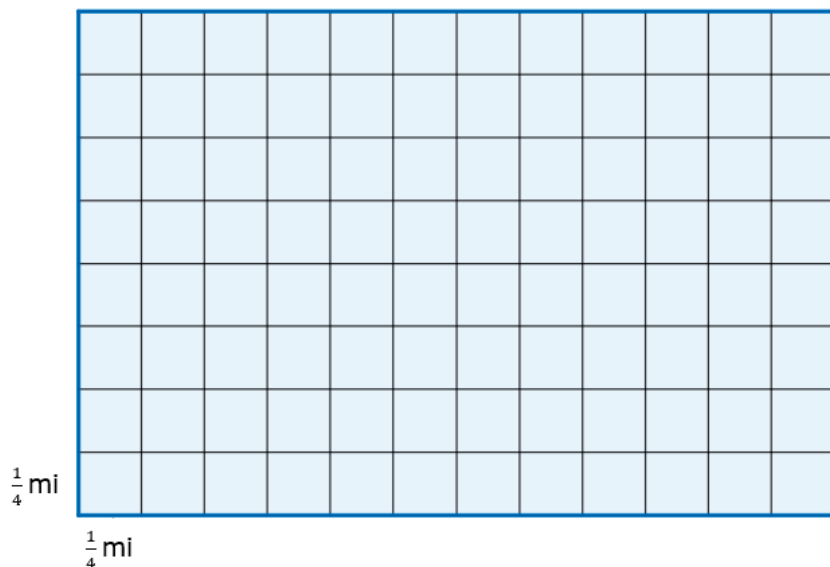
9.2.4 Guided Practice: Density Based on Area

1. The number of leaves on a tree can be estimated using the crown of the tree. A tree crown is the top part of the tree where branches grow out from the main trunk. The width of the tree's crown is used as the diameter of the circle of the ground that lies under the tree's crown. The area where the leaves could fall is 4 times the area of the circle that lays under the tree's crown. The approximate number of leaves per square foot is based on the average size of a leaf for each type of tree. Use the table to approximate the number of leaves that could fall from each tree.

Tree	Crown (in feet)	Approximate Number of Leaves per Square Foot
Sycamore	35	5
Live Oak	16	14

2. The population density of Gainesville, Florida, is 2028.4 persons per square mile, and the population of Gainesville is 124,504. What is the area, in square miles, of Gainesville?

3. Wildlife biologists divided a habitat in the Bear Island area of the Big Cypress National Preserve into 96 squares, as shown. Each square has the dimensions of $\frac{1}{4}$ mile (mi) by $\frac{1}{4}$ mi.



The biologists found that in the Bear Island habitat, there are 13.8 deer per square mile.

- What is the population density for one of the $\frac{1}{4}$ mi by $\frac{1}{4}$ mi square?
- How many deer live in the habitat area the biologists studied?



9.2.5 Wrap-Up: Cool Down

- Write an explanation to a classmate who might have missed today's lesson, in your own words, about population density.